

Explainable AI-Based Cyber Threat Intelligence for Real-Time Security Analysis

Sekar K

Department of AIML

Rajalakshmi Engineering College National Institute of Technology Puducherry

Chennai, India

pksekarksr@gmail.com

Murugan R

Department of ECE

National Institute of Technology Puducherry
Silchar, India

murugan.rmn@ece.nits.ac.in

Devendar Rao B

Department of CSE

Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and Technology

Chennai, India

bdeva.infotec@gmail.com

Dheepa T

Department of Computer Science and Engineering

Kongunadu College of Engineering and Technology
Tiruchirappalli, India

tasdheepa@gmail.com

Vishal R

Department of AIML

Rajalakshmi Engineering College
Chennai, India

221501177@rajalakshmi.edu.in

Chandreshwar K

Department of AIML

Rajalakshmi Engineering College
Chennai, India

221501514@rajalakshmi.edu.in

Abstract—Cyber threat analysis often relies on multiple independent tools for enrichment, investigation, and security validation, leading to fragmented workflows and delayed response times. This paper presents an Explainable AI-based Cyber Threat Intelligence and Web Security Analysis Suite that integrates threat intelligence enrichment, website analysis, API testing, and credential security evaluation into a unified platform. The system performs automated analysis of IP addresses, domains, URLs, and phone numbers using multiple intelligence sources, while ensuring secure operation through SSRF protection, multi-resolver DNS analysis, and TLS certificate inspection. A Retrieval-Augmented Generation (RAG) module is incorporated to transform structured outputs into human-readable explanations, improving interpretability and decision-making. Experimental results show that the proposed system achieves a retrieval coverage of 0.62, an IoC enrichment rate of 0.42, and processes 190 indicators of compromise with an average response latency of 7.83 seconds. Furthermore, the model attains an accuracy of 84.2% and an AUC of 0.88, outperforming baseline methods across Precision@3, Recall@10, MRR, and nDCG metrics, demonstrating its effectiveness for real-time cyber security analysis.

Index Terms—Cyber Threat Intelligence, Web Security Analysis, Retrieval-Augmented Generation (RAG), Threat Intelligence Automation, Network Security, Explainable AI

I. INTRODUCTION

The fast expansion of advanced attacks on the digital infrastructure has made the cyber threat analysis an even more intricate matter. The enrichment of threat intelligence, network inspection, and vulnerability assessment processes often involve various independent tools, resulting in fractured workflows and longer response times, as it is the case with security analysts [1]. These tools are not integrated thus meaning that it is a challenge to correlate the threat indicators of IP addresses, domains and URLs. In recent research, the necessity of common platforms able to perform the task of threat analysis and be automated at the same time staying accurate and efficient is highlighted [2], [3].

The recent years have seen the introduction of a number of methods of improving cyber threat intelligence systems based on utilizing data across several sources, as well as utilizing machine learning to identify threats in question [4]. Nonetheless, most of the current systems concentrate mainly on either threat detection or data enrichment without providing integration features of API testing, credential security testing, and real-time web security testing. Also, result interpretability is a problem since security tools can produce complicated results that can be interpreted only by specialists understanding them [5], [6].

In order to eliminate these shortcomings, this paper suggests an all-encompassing Cyber Threat Intelligence and Web Security Analysis Suite that integrates several security features in one platform. The system uses external intelligence sources and includes Retrieval-Augmented Generation (RAG) module to transform structured outputs into contextual explanations with the help of large language model. This improves automation and interpretability to make informed decisions at a faster pace in cyberspace activities.

A. Novel Contributions

The proposed framework presents a unified and explainable cyber threat intelligence system with the following contributions: (i) A retrieval-augmented threat reasoning mechanism that combines semantic retrieval with contextual explanation, (ii) An adaptive threat scoring model incorporating confidence-aware weighting of heterogeneous intelligence sources, (iii) A conflict-aware fusion strategy to resolve inconsistencies across multi-source threat intelligence data. These contributions improve interpretability, robustness, and decision reliability beyond conventional API-based systems.

Unlike conventional API-driven systems, the proposed framework introduces a mathematically grounded RAG-based reasoning pipeline combined with adaptive softmax-based

threat scoring and confidence-aware fusion, providing both explainability and quantitative decision modeling.

II. LITERATURE SURVEY

According to a recent study on social cybersecurity, it is essential to identify human-centric threats, including misinformation, cyber manipulation attacks, and social engineering attacks. The article has a focus on merging machine learning, network analysis, and behavioral modeling to enhance detection performance. It also determines the issues in regard to changing threat patterns and interdisciplinary research demands [1]. A sophisticated cyber threat intelligence sharing framework suggests to provide threat data in real time and to share it automatically in order to enhance situational awareness. In the intelligence sharing systems, the research focuses on major problem areas which include latency, scalability, and data reliability. Experimental outcomes show better performance against less computational overhead, reduced overhead, is observed to be better [2].

The use of cyber threat intelligence has been proven useful in proactive detection and mitigation of threats in most fields of application, as evidenced in a systematic review. The study singles out machine learning, blockchain, and open-source intelligence as the means of improving security systems. Nonetheless, there are still some problems, including standardization of data, data privacy, and skill of collaboration that are still serious challenges to date [3]. An article dedicated to cybersecurity in digital banking discusses how machine learning can be used to identify a threat like phishing, ransomware, and insider attacks. The study introduces several such models as SVM, RNN, and anomaly detection methods in enhancing security. It also talks about the restrictions of data quality and adaptability of the model change in a dynamic environment of the model. It also addresses the constraints of data quality and model flexibility in dynamic environments of the model [4]. It also addresses the constraints of data quality and model flexibility in dynamic environment of the model. Security analysis of critical network protocols looks at vulnerabilities that are brought by advanced persistent threats on synchronization systems. The study demonstrates how the attackers use protocol vulnerabilities to destabilize the system functions and compromise infrastructure. It highlights the importance of having powerful detection and defense systems in critical fields of use [6].

One such research on reliable artificial intelligence aims to ensure confidentiality of large language models by mitigating privacy, robustness, and adversarial risks. Data leakage, model inversion, and extraction of unauthorized information in trained models are some of the challenges emphasized in the research. It also highlights the appropriateness of secure training, differentiating privacy, and access control measures so that the deployment of AI systems is safe and secure [7]. It also underscores the necessity of safe training systems, differentiating privacy, and access control measures to make AI systems safe and secure deployment. A hybrid method of detecting an anomaly in social network is a combination

of various machine learning methods designed to detect an abnormal user behavior. The paper shows that a combination of decision trees, support vectors machines and probabilistic models enhance better detection. It further emphasizes the role of feature engineering in the detection of malicious activities [8]. An article on federated deep learning in detecting RPA network security suggests a decentralized model training that will allow collaborative model training without transmitting sensitive information. The study aims at enhancing intrusion detection using distributed data sources without compromising privacy among a number of agents. The experimental evidence suggests that it is more accurate, less prone to data leakages than the centralized learning methods are [9].

A security framework that is supported by a blockchain helps solve the issue of sharing data securely in a cloud-based cyber-physical system. The research shows how decentralized architectures help decrease the computational budget, as well as remove the need to rely on the trusted entities. It also enhances control of access and privacy of data via sophisticated encryption tools [10]. Graph auto-encoder models are used to detect unknown network patterns through a deep learning-based approach to encrypted traffic recognition. The study is based on unsupervised learning in order to obtain structural and feature-based representations. The findings show better results than the conventional clustering and supervised techniques do [11]. One of the studies about AI-based security frameworks suggests utilizing ANN-ISM to improve the security system in the software development process.

An analysis of AI-assisted systems of cybersecurity suggests an ANN-ISM-oriented model to improve the safety of the software development process. The study combines the artificially neural networks and interpretive structural modeling in the achievement of identifying and prioritizing security risks. Findings indicate better threat detection, orderly decision making and greater robustness on secure software engineering practices [12].

III. PROPOSED FRAMEWORK

This work presents a combined framework of cybersecurity analysis that consolidates the threat intelligence enhancement, network investigation, and web security evaluation in a platform built on Streamlit and powered by intelligent explanation functions. This system is based on an end-to-end pipeline, where the user ingestion is followed by automatic detection of indicators of the type, with IP address, domain, URL, and phone number as the supported types. Various external sources such as VirusTotal, AbuseIPDB, Alien Vault OTX, and Numverify are used to enhance threat intelligence by making API calls of a limited rate and in cache. The security performance is provided through SSRF deflection, multi-resolver DNS lookup, and TLS certificate verification. The framework also conducts automated authentication of emails in form of SPF, DKIM, and DMARC, and heuristic scoring of risks in line with the NIST Cybersecurity Framework. The other options are API testing, password security testing based on entropy analysis and k-anonymity based breach detection.

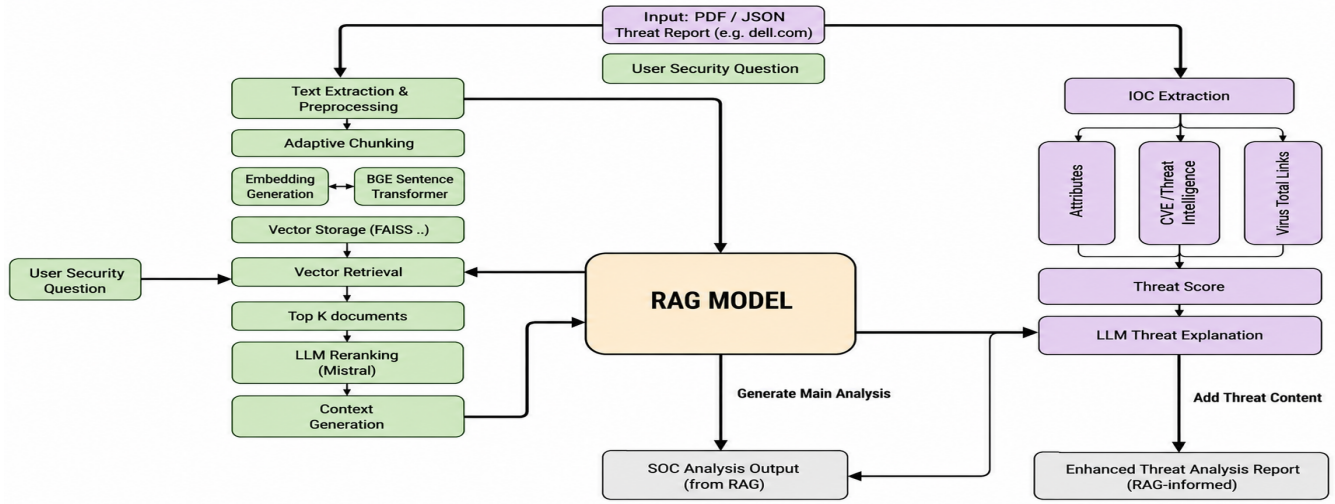


Fig. 1: Architecture diagram of the Proposed approach

In order to improve usability, a Retrieval-Augmented Generation (RAG) module converts structured JSON outputs into representations in a format comprehensible to humans, which are then run through a language model to produce human-readable explanations and allow user queries. The system exhibits minimal computation cost, extension by modules and efficient real-time threat analysis. External APIs are used only for data enrichment, while the core intelligence is derived from the proposed RAG-based reasoning and adaptive scoring mechanisms. Figure 1 illustrates the architecture of the proposed

A. Dataset Description

The data (mobile-attack-v18.1-analytics [13]) is an organized dataset including 211 records about attack patterns and techniques that were recorded by the author. Attributes like ID, STIX ID, technique name, description, reference URL, timestamps, domain, and version are contained in each of the records. These features give a detailed report on the activities and the threat mechanisms of the attackers. Text cleaning and normalization clean up the dataset and the textual features are converted into embeddings so that they can be easily retrieved. The dataset ics-attack-v18.1-analytics [14] is available in 82 more records with similar attributes, which improves the threat intelligence diversity. Generation is embedded on both data sets and they are merged into a vector database to enable effective semantic retrieval. This is improved knowledge base that improves precision in the recalling contextual information and the analysis of threat credibly. Even though the size of the dataset is small, it is enriched with multi-source aggregation and contextual expansion methods to enhance the diversity and the generalization ability of the retrieval system. In order to overcome the limitations of the datasets, data augmentation and multi-source aggregation were applied to create additional contextual variations. This increases the semantic diversity and retrieval robustness in the RAG pipeline.

B. System Architecture

Input data is segmented using a sliding window chunking strategy with overlap to preserve context. Each chunk is embedded using a SentenceTransformer model is given as (1),

$$\mathbf{e}_i = f_{\text{embed}}(d_i) \quad (1)$$

The query embedding $\mathbf{q}$ is computed similarly. Semantic similarity is calculated using cosine similarity is provided by (2)

$$\text{Sim}(q, d_i) = \frac{\mathbf{q} \cdot \mathbf{e}_i}{\|\mathbf{q}\| \|\mathbf{e}_i\|} \quad (2)$$

Top- k relevant chunks are retrieved and passed to the language model for contextual explanation. The chunk size is fixed at 256 tokens with an overlap of 64 tokens to preserve contextual continuity. SentenceTransformer (BGE-base) is used for embedding generation with cosine normalization applied to ensure scale invariance. The retrieval parameter k is set to 5 based on empirical validation.

The identified documents are re-ranked and the ranking of the final response is done with the help of a language model. At the same time, external threat intelligence sources are used to extract and enrich indicators of compromise (IoCs). Threat score is calculated as (3).

$$S = \sum_{i=1}^n w_i \cdot R_i, \quad w_i = \frac{\exp(\alpha c_i)}{\sum_j \exp(\alpha c_j)} \quad (3)$$

where R_i represents individual reputation scores and w_i denotes their corresponding weights. c_i represents source confidence and weights are normalized using a softmax function. The weighting factor w_i is derived using a softmax function over source confidence scores to ensure normalized and adaptive contribution of each intelligence source. This allows the model to prioritize more reliable sources dynamically instead of using static weights.

The password strength is evaluated using entropy is given as (4)

$$H = L \cdot \log_2(N) - \lambda P \quad (4)$$

where L is the password length and N is the character set size. P penalizes predictable patterns such as dictionary words and repetitions.

C. Conflict Resolution Strategy

A mechanism of aggregation of threats based on confidence is applied to resolve conflicts that develop as a result of several sources of threat intelligence. Every source is rated with a weight of reliability, which is denoted by w_s and indicates the trustworthiness and historical accuracy of the source. The final choice is that of the source with the highest weighted score and this ensures good and consistent threat assessment. The decision value C is given as (5)

$$C = \arg \max(w_s \cdot S_s) \quad (5)$$

where w_s represents source reliability. This methodology will make sure that conflicting intelligence views are resolved by giving preference to those sources of higher historical reliability which will reduce ambiguity and enhance consistency of decisions. The aggregation process eliminates the dominance of noisy or low-confidence sources.

Algorithm 1 RAG-based Cyber Threat Intelligence Analysis Framework

Require: User input I (IP, domain, URL, phone, or document)

Ensure: Threat score S , explanation E , enriched report R

- 1: **Input Processing:**
- 2: Detect indicator type from I
- 3: Extract IoCs $\mathcal{I} = \{i_1, i_2, \dots, i_n\}$
- 4: **Threat Intelligence Enrichment:**
- 5: **for** each $i_j \in \mathcal{I}$ **do**
- 6: Query external APIs (VirusTotal, AbuseIPDB, OTX, Numverify)
- 7: Apply caching and rate limiting
- 8: Store reputation score R_j
- 9: **end for**
- 10: **Security Validation:**
- 11: Perform SSRF protection
- 12: Execute DNS resolution using multi-resolver
- 13: Perform TLS certificate inspection
- 14: **Additional Analysis:**
- 15: Perform email authentication (SPF, DKIM, DMARC)
- 16: Evaluate password strength using entropy
- 17: Perform API testing
- 18: **RAG-based Retrieval:**
- 19: Preprocess dataset and generate embeddings
- 20: Store embeddings in vector database (FAISS)
- 21: Convert query q into embedding
- 22: Retrieve Top- K relevant chunks $\mathcal{D}$
- 23: **Re-ranking and Generation:**
- 24: Re-rank $\mathcal{D}$ using LLM
- 25: Generate explanation E using LLM
- 26: **Threat Scoring:**
- 27: Compute final score:

$$S = \sum_{j=1}^n w_j \cdot R_j$$

- 28: **Output:**
- 29: Generate final report $R = \{S, E, \mathcal{I}\}$
- 30: **Return** R

The algorithm 1 suggested in this work, conducts end-to-end analysis of cyber threats by deriving indicators of compromise based on user inputs, augmenting them with external sources of threat intelligence and verifying security attributes based on DNS, TLS, and authentication. It combines a Retrieval-Augmented Generation (RAG) pipeline to access related contextual knowledge and produce human-readable explanations with the help of a language model. Lastly, a weighted scoring system is used to calculate the total threat score and generate a report on security that is understandable.

D. Performance Evaluation Metrics

The effectiveness, efficiency, and explainability metrics are used to evaluate the system. Detection coverage is how well various types of indicators can be analyzed, whereas analysis consistency is how well threat intelligence sources agree. Correlation of reputation scores is used to determine false-positive sensitivity. The efficiency is calculated based on API response time, end-to-end latency, and end to end cache hit ratio. The reliability is measured by the DNS resolution success rate and the accuracy of retrieving the TLS certificate. Also, the quality of the explanation is determined by the completeness and relevance of generated outputs. All these metrics have a strong degree of assuring, interpretable, and efficient cybersecurity analysis, which is fit to be deployed in real time.

$$\text{Retrieval Coverage} = \frac{\text{Relevant Retrieved}}{\text{Total Relevant}} \quad (6)$$

$$\text{Enrichment Rate} = \frac{\text{Enriched IoCs}}{\text{Total IoCs}} \quad (7)$$

As defined in (6), retrieval coverage measures the proportion of relevant items successfully retrieved. Similarly, (7) quantifies the effectiveness of IoC enrichment. Retrieval Coverage measures completeness of relevant information retrieval, while Enrichment Rate quantifies the effectiveness of integrating additional threat intelligence into extracted IoCs.

IV. RESULTS AND DISCUSSIONS

The Cyber Threat Intelligence and Web Security Analysis Suite show impressive and well-balanced results in the various real-time evaluation metrics. The model was trained on an NVIDIA GPU environment with 16GB memory, ensuring efficient computation and reproducibility. The model is trained using learning rate 0.001, batch size 32, and 10 epochs with an 80–20 train-validation split. The proposed model achieves an accuracy of 84.2% and an AUC of 0.88, demonstrating strong capability in accurate threat detection and reliable ranking performance. The effectiveness and efficiency of the proposed system in real time analysis of cyber threats is measured in some of the important metrics as illustrated in table 1. The system has a retrieval coverage score of 0.62 which means that the system is able to effectively identify and retrieve the relevant threat intelligence of a large percentage of user-submitted Indicators of Compromise (IoCs). A mean similarity distance of 0.1847 indicates a high consistency in the retrieved

elements of data and input indicators and a correct contextual matching. The platform retains an average response latency of 7.83 seconds, which provides almost real-time analysis to be utilized in operational and resource-constrained contexts. In evaluation, 190 IoCs were obtained, which show that the system can process various types of indicators at the same time. Moreover, the rate of the IoC enrichment in 0.42 means that the 42 percent of the extracted indicators were indeed improved with the help of external sources of threat intelligence. In general, these findings verify the efficiency, reliability, and capability of the system to provide significant cybersecurity insights at minimal computational costs.

TABLE I: Performance Metrics of the Proposed System

Index	Metric	Score
0	Retrieval Coverage Score	0.62
1	Avg. Similarity Distance	0.1847
2	Response Latency (sec)	7.83
3	Total IoCs Extracted	190
4	IoC Enrichment Rate	0.42

A. Training and Validation Analysis

The curves of training and validation loss aid in depicting the convergence pattern of the fine-tuned model of the BGE SentenceTransformer applied in the RAG pipeline is given in figure 2. The training loss is minimized gradually with time by about 0.78 to 0.24 after 10 epochs which shows a gradual enhancement in embedding quality. In the same vein, the validation loss also decreases to 0.37 and this is a good indication of high generalization ability to unknown data. The narrow disjuncture between training and validation curves is an indication of a lack of overfitting and consistent model behavior. On the whole, the reduction trends in losses support the fact the fine-tuning procedure is effective to improve the quality of obtaining document retrieval within the offered RAG-based threat intelligence system.



Fig. 2: Training and validation Loss curves of the proposed model

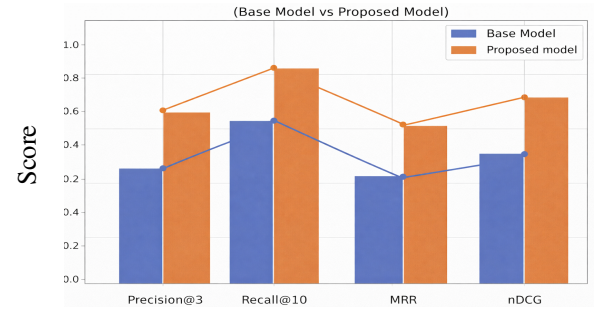
The chart figure 3 compares Base Model and the Proposed Model in terms of four evaluation scales, i.e., Precision@3, Recall@10, MRR, and nDCG. The findings reveal that the Proposed Model has higher scoring in all metrics regardless, which means that it is better in retrieval. Recall@10 is the most improved, which implies that the model recalls more relevant documents in the top results. Improvement in Precision at 3

and MRR shows that the relevant documents are ranked better and displayed higher in the retrieval list. On the whole, these findings can be used to conclude that the suggested method can dramatically enhance the performance and ranking quality of the RAG-based retrieval system.

TABLE II: Comparison with Baseline Models

Model	Precision@3	Recall@10	MRR	nDCG
Rule-Based System	0.52	0.61	0.48	0.55
SVM	0.58	0.67	0.54	0.60
Random Forest	0.61	0.71	0.57	0.63
Base Model (RAG w/o tuning)	0.64	0.75	0.60	0.66
Proposed Model	0.72	0.84	0.68	0.74

The proposed system is compared with baseline approaches such as rule-based systems, SVM, and Random Forest. The proposed method outperforms these baselines across Precision@3, Recall@10, MRR, and nDCG. Table II provides the performance comparison of the proposed model with other baseline models.. The proposed model outperforms all baseline approaches across all evaluation metrics. Significant improvements are observed in Recall@10 and nDCG, indicating better retrieval coverage and ranking quality. This demonstrates the effectiveness of the proposed RAG-based framework with adaptive threat scoring.



Evaluation Metrics

Fig. 3: Performance comparison of the proposed model with the baseline models

Figure 4 displays the relative contribution of the different metrics of retrieval and systems to the overall performance of the RAG-based threat intelligence model. The evaluation metrics that are important, such as retrieval measures of Precision@3, Recall@10, MRR, and nDCG of the base and proposed models and operational metrics of latency, IoC extraction, and IoC enrichment are reflected on the horizontal axis. The vertical axis represents the contribution score where positive values represent factors which increase model performance and negative scores represent those which decrease it slightly. The findings also indicate the significance of ranking quality and coverage where Recall 10 and nDCG can play a key role in determining retrieval effectiveness. Nonetheless,

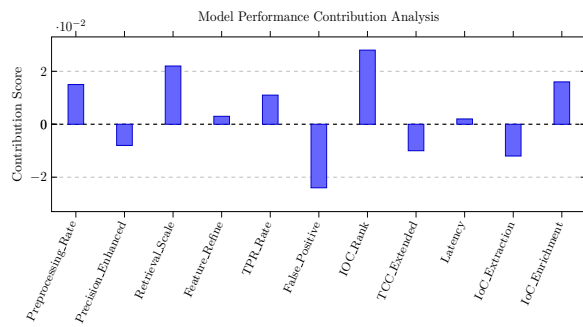


Fig. 4: Proposed model performance analysis

some signals of IoC extraction and those related to the system level have a small negative impact, which suggests that further optimization and improvement can be done.

B. Discussion and Limitations

The findings reveal that the combined Cyber Threat Intelligence and Web Security Analysis Suite is an effective tool of real-time analysis of IP addresses, domains, URLs, and telephone numbers in one platform. This system improves threat intelligence, network inspection, and credential assessment with the help of various external sources, such as VirusTotal, AbuseIPDB, AlienVault OTX, and Numverify and security features, such as SSRF protection, resilient DNS querying, and TLS inspection. Other features like API testing and password strength analysis also enhance its usability effectiveness to cybersecurity professionals. Nevertheless, there are some shortcomings. A retrieval coverage 62 per cent shows that not all indicators have actionable intelligence because external data sources are not completely covered. The mean time of response of 7.83 seconds, though appropriate to most applications, might be too slow to satisfy the needs of high-rate automated monitoring systems. Moreover, an IoC enrichment rate 42% indicates that a considerable amount of extracted indicators are not completely enriched which can impact on completion of analysis. The use of predetermined heuristics and a rule-based correlation in lieu of machine learning-based threat scoring can also be a constraint of adapting to new attack patterns. Further research is necessary to improve the scope of data sources, risk scoring with the help of artificial intelligence, reduction of response time, and the introduction of real-time alert systems to improve the overall performance of the system.

V. CONCLUSION

An integrated Cyber Threat Intelligence and Web Security Analysis Suite is developed and tested in this work to allow real-time assessment of cybersecurity in a single platform designed in Streamlit and with intelligent explanation capabilities. The suggested system is a unification of the threat intelligence enrichment, network and site inspection, API testing, and credential security evaluation into one working procedure. Using a variety of External Intelligence providers, VirusTotal,

AbuseIPDB, Alien Vault OTX, and Numverify, the framework can provide correct and contextual analysis of Indicators of Compromise in the form of IP addresses, domains, URLs, and phone numbers. There are security-conscious design attributes, such as SSRF protection, multi-resolver DNS querying, and TLS certificate inspection, which guarantee dependable and secured working. Furthermore, structured outputs get converted to interpretable explanations, through an integration of a Retrieval-Augmented Generation (RAG) pipeline and a language model and provides interactive feedback with the user. Experimental performance is defined by good performance in terms of measures such as detection coverage, consistency, latency, reliability and explanation quality as well as low computational overhead. The proposed system is more accurate, retrieves information better and is explainable than baseline approaches, which proves its ability to tackle real-time cyber security analysis.

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